STAT 559 – Note 7
Modes of Convergence and Uniform Integrability

Edited transcription of handwritten lecture notes

These notes transcribe and edit the indicated handwritten PDF. Source-page markers preserve traceability. Yellow boxes retain original TODOs; red boxes explain mathematical corrections; blue boxes add explanations or proof details. The reference is *stats310notes.pdf* (provided in `notes/`); references below use its printed section numbers.

Source page(s): 1–3

1 Four modes of convergence

Unless otherwise specified, X_n and X are random variables on a common probability space $(\Omega, \mathcal{F}, \mathbb{P})$. *Almost sure convergence*, written $X_n \to X$ a.s., means

$$\mathbb{P}(\{\omega : \lim_n X_n(\omega) = X(\omega)\}) = 1. \tag{1}$$

Convergence in probability, written $X_n \xrightarrow{\mathbb{P}} X$, means

$$\mathbb{P}(|X_n - X| > \varepsilon) \longrightarrow 0 \quad \text{for every } \varepsilon > 0. \tag{2}$$

Convergence in L^p, for $1 \le p \le \infty$, means $X_n, X \in L^p$ and

$$\|X_n - X\|_p \longrightarrow 0. \tag{3}$$

For finite p this is equivalently $\mathbb{E}|X_n - X|^p \to 0$; $p = 1$ is convergence in mean. *Convergence in distribution*, also called convergence in law or weak convergence and written $X_n \Rightarrow X$ or $X_n \xrightarrow{d} X$, means

$$F_{X_n}(t) \longrightarrow F_X(t) \quad \text{at every continuity point } t \text{ of } F_X. \tag{4}$$

For this last notion the variables may live on different probability spaces; only their laws are compared. Convergence is not required at discontinuities of the limiting cdf (reference Section 8.1).

Correction: Which measure tests almost sure convergence?

The first page writes a null-set expression using μ_X. The convergence event is a subset of the sample space Ω, so it is measured by $\mathbb{P}$, not by the law μ_X on $\mathbb{R}$.

Source page(s): 4–5

2 Almost sure convergence implies convergence in probability

Fix $\varepsilon > 0$ and put $S_n = \bigcap_{k \geq n} \{|X_k - X| \leq \varepsilon\}$. Almost sure convergence implies $\mathbb{P}(\bigcup_n S_n) = 1$. Since S_n increases and $S_n \subseteq \{|X_n - X| \leq \varepsilon\}$,

$$1 = \lim_n \mathbb{P}(S_n) \leq \liminf_n \mathbb{P}(|X_n - X| \leq \varepsilon) \leq 1. \tag{5}$$

The complement probabilities tend to zero.

Original TODO: Translate the limit statement into events

"Eventually within ε" means "there exists n such that for every $k \geq n$ the bound holds." The existential quantifier becomes a union and the universal quantifier becomes an intersection. To express convergence itself, use a countable family of tolerances such as $1/m$ and intersect over m. Interchanging an uncountable family of probability-one events would require care; the countable tolerances avoid that issue (reference Proposition 8.2.1).

Source page(s): 5–7

3 Convergence in probability has an a.s. convergent subsequence

If $X_n \xrightarrow{\mathbb{P}} X$, choose increasing indices n_k such that

$$\mathbb{P}(|X_{n_k} - X| > 2^{-k-1}) \leq 2^{-k}. \tag{6}$$

The thresholds decrease, so both n_k and n_{k+1} can be chosen late enough for the bound at tolerance 2^{-k-1}. The triangle inequality gives

$$\mathbb{P}(|X_{n_k} - X_{n_{k+1}}| > 2^{-k}) \leq 2 \cdot 2^{-k}. \tag{7}$$

The constant factor does not affect summability. Borel–Cantelli and telescoping, as in the L^p completeness proof, show that X_{n_k} is pointwise Cauchy outside a null set. It converges there to a measurable Y; set $Y = 0$ on the exceptional set. It remains to identify $Y = X$ a.s.

Original TODO: Final-exam candidate: uniqueness of the probability limit

Almost sure convergence to Y implies convergence in probability to Y. For any $\varepsilon > 0$,

$$\mathbb{P}(|X - Y| > \varepsilon) \leq \mathbb{P}(|X - X_{n_k}| > \varepsilon/2) + \mathbb{P}(|X_{n_k} - Y| > \varepsilon/2) \longrightarrow 0. \tag{8}$$

Taking $\varepsilon = 1/m$ for all positive integers m proves $X = Y$ a.s. Thus the subsequence converges a.s. to X (reference Proposition 8.2.3).

Supplement: The source's Riesz–Fischer connection

The source reuses the Cauchy-subsequence technique from the Riesz–Fischer proof. The result here is the subsequence principle for convergence in probability; Riesz–Fischer itself states completeness of L^p. A handwritten reminder "fill in: 12:12 pm" appears beside the proof, but

no corresponding recording is supplied. The complete argument above covers the mathematical gap. One can also prove the claim directly by Borel–Cantelli on $\{|X_{n_k} - X| > 2^{-k}\}$.

Source page(s): 8–9

4 Convergence in probability implies convergence in law

For $\varepsilon > 0$ the event inclusions give both bounds

$$F_{X_n}(t) \le F_X(t+\varepsilon) + \mathbb{P}(|X_n - X| > \varepsilon), \tag{9}$$
$$F_{X_n}(t) \ge F_X(t-\varepsilon) - \mathbb{P}(|X_n - X| > \varepsilon). \tag{10}$$

The first follows because $X_n \le t$ and $|X_n - X| \le \varepsilon$ imply $X \le t + \varepsilon$. The second follows because $X \le t - \varepsilon$ and the same closeness imply $X_n \le t$. Taking limsup and liminf in n yields

$$F_X(t-\varepsilon) \le \liminf_n F_{X_n}(t) \le \limsup_n F_{X_n}(t) \le F_X(t+\varepsilon). \tag{11}$$

At a continuity point t let $\varepsilon \downarrow 0$; the outer terms both tend to $F_X(t)$, proving the result (reference Proposition 8.2.4).

Correction: The lower-bound step

The source says to obtain the other bound by choosing a negative epsilon. The error tolerance remains positive; the separate inclusion using $t - \varepsilon$ above gives the lower bound. Continuity at t, rather than right-continuity alone, is needed on that side.

Source page(s): 10

5 Slutsky's theorem and asymptotic linearity

If $X_n \Rightarrow X$ and $Y_n \xrightarrow{\mathbb{P}} c$ for a constant $c \in \mathbb{R}$, then

$$X_n + Y_n \Rightarrow X + c, \qquad X_n Y_n \Rightarrow cX. \tag{12}$$

For each n the pair X_n, Y_n must be defined together; the limit X need not share their probability space. Independence is not required.

Supplement: The assigned Slutsky proof

For the sum, repeat the preceding cdf bounds comparing $X_n + Y_n$ to $X_n + c$. Choose decreasing positive tolerances for which the shifted arguments are continuity points of the limiting cdf; their complement is countable, so this is possible. Then let the tolerance tend to zero. For the product, $X_n \Rightarrow X$ implies that the sequence is tight: given $\eta > 0$, choose large continuity points $-M, M$ with limiting tail probability below η. Cdf convergence then bounds the tails of X_n for all sufficiently large n. Thus

$$\mathbb{P}(|X_n(Y_n - c)| > \varepsilon) \le \mathbb{P}(|X_n| > M) + \mathbb{P}(|Y_n - c| > \varepsilon/M) \tag{13}$$

tends to at most η, and hence tends to zero after $\eta \downarrow 0$. Since $cX_n \Rightarrow cX$ (by scaling, or trivially for $c = 0$), the sum case applied to $cX_n + X_n(Y_n - c)$ proves the product case. See reference Proposition 8.2.7.

An asymptotically linear estimator has an expansion

$$\widehat{\theta}_n = \theta_0 + \frac{1}{n}\sum_{i=1}^{n} Z_i + R_n, \qquad \sqrt{n}R_n \xrightarrow{\mathbb{P}} 0. \tag{14}$$

If the leading sum satisfies a CLT, for example when Z_i are iid with mean zero and finite variance σ^2, then

$$\sqrt{n}(\widehat{\theta}_n - \theta_0) = \frac{1}{\sqrt{n}}\sum_{i=1}^{n} Z_i + \sqrt{n}R_n \Rightarrow N(0, \sigma^2). \tag{15}$$

Supplement: The assumptions behind the statistical example

The source labels the leading term normal by the CLT without stating assumptions on Z_i. The iid mean-zero finite-variance conditions above are one sufficient set; asymptotic linearity alone does not establish the CLT.

Source page(s): 11–12

6 Lp convergence and bounded convergence in probability

For finite $p \geq 1$, Markov gives

$$\mathbb{P}(|X_n - X| > \varepsilon) \leq \frac{\mathbb{E}|X_n - X|^p}{\varepsilon^p} \longrightarrow 0 \tag{16}$$

when $X_n \to X$ in L^p. For $p = \infty$, essential-supremum convergence gives the same conclusion directly. Conversely, suppose $X_n \xrightarrow{\mathbb{P}} X$ and $|X_n| \leq C < \infty$ a.s. for one common deterministic constant. Then $X_n \to X$ in every finite L^p.

Original TODO: Verify the bound for the limit and finish the truncation argument

For any $a > 0$,

$$\mathbb{P}(|X| > C + a) \leq \mathbb{P}(|X - X_n| > a) \longrightarrow 0. \tag{17}$$

A countable union over $a = 1/m$ gives $|X| \leq C$ a.s. For any $\delta > 0$ split the expectation according to $|X_n - X|^p > \delta$:

$$\mathbb{E}|X_n - X|^p \leq (2C)^p \mathbb{P}(|X_n - X| > \delta^{1/p}) + \delta. \tag{18}$$

Therefore $\limsup_n \mathbb{E}|X_n - X|^p \leq \delta$. Let $\delta \downarrow 0$. Use limsup here because the existence of the expectation limit has not yet been proved (reference Proposition 8.2.9).

The implication diagram in the source is therefore

$$\text{a.s. convergence} \implies \text{convergence in probability} \implies \text{convergence in law}, \tag{19}$$

with L^p convergence also implying convergence in probability. Reverse implications require additional assumptions; convergence in probability always supplies an a.s. convergent subsequence. If a law limit is a constant c, then $X_n \Rightarrow c$ does imply $X_n \xrightarrow{\mathbb{P}} c$, as shown by cdf bounds at $c \pm \varepsilon/2$ (reference Exercise 8.2.6).

Source page(s): 13–14

7 Uniform integrability

A family $\{X_n\}$ is *uniformly integrable* (UI) if

$$\lim_{K\to\infty} \sup_n \mathbb{E}\big[|X_n|\mathbf{1}_{\{|X_n|>K\}}\big] = 0. \tag{20}$$

Equivalently, for every $\varepsilon > 0$ a single cutoff K makes the tail expectation at most ε for every n. A sufficient criterion is the existence of $r > 1$ such that $M_r := \sup_n \mathbb{E}|X_n|^r < \infty$. On $\{|X_n| > K\}$,

$$|X_n| = \frac{|X_n|^r}{|X_n|^{r-1}} \le \frac{|X_n|^r}{K^{r-1}}, \tag{21}$$

so

$$\sup_n \mathbb{E}\big[|X_n|\mathbf{1}_{\{|X_n|>K\}}\big] \le \frac{M_r}{K^{r-1}} \longrightarrow 0. \tag{22}$$

Original TODO: Why does taking a supremum express uniformity?

The finite number M_r bounds the moment of every variable simultaneously. Consequently the same choice $K > (M_r/\varepsilon)^{1/(r-1)}$ works for all n. Separate finite moments $\mathbb{E}|X_n|^r < \infty$, without a common bound, would not provide that conclusion. This also explains the denominator comparison marked for review on page 14 (reference Proposition 8.3.2).

Source page(s): 15–17

8 Vitali's L1 convergence theorem

If $X_n \xrightarrow{\mathbb{P}} X$ and $\{X_n\}$ is UI, then X is integrable and $\mathbb{E}|X_n - X| \to 0$. In particular this holds under a.s. convergence. The source proves the probability version by truncation; reference Proposition 8.3.1 states the a.s. version. Define the clipping function

$$\phi_K(x) = \max(-K, \min(x, K)). \tag{23}$$

It is continuous, bounded, and $|x - \phi_K(x)| \le |x|\mathbf{1}_{\{|x|>K\}}$.

Supplement: Continuous mapping used in the proof

If $g : \mathbb{R} \to \mathbb{R}$ is continuous and $X_n \xrightarrow{\mathbb{P}} X$, then $g(X_n) \xrightarrow{\mathbb{P}} g(X)$. For a direct argument, restrict to $|X| \le M$ with high probability, use uniform continuity of g on $[-M-1, M+1]$, and bound the remaining error by $\mathbb{P}(|X| > M) + \mathbb{P}(|X_n - X| > \delta)$. Let $n \to \infty$ and then $M \to \infty$. Almost sure convergence is preserved by pointwise continuity. The distribution version follows from the continuous mapping theorem (reference Section 8.7).

By the bounded-convergence-in-probability result, for fixed K,

$$\mathbb{E}|\phi_K(X_n) - \phi_K(X)| \longrightarrow 0. \tag{24}$$

UI gives $\sup_n \mathbb{E}|X_n| < \infty$ (proved below). Take an a.s. convergent subsequence and apply Fatou to obtain $\mathbb{E}|X| < \infty$. Then DCT gives $\mathbb{E}|X - \phi_K(X)| \to 0$. Finally,

$$\mathbb{E}|X_n - X| \le \mathbb{E}|X_n - \phi_K(X_n)| \tag{25}$$

$$+ \mathbb{E}|\phi_K(X_n) - \phi_K(X)| + \mathbb{E}|\phi_K(X) - X|. \tag{26}$$

Original TODO: Make the epsilon argument rigorous; why is X integrable?

Given $\varepsilon > 0$, choose K so the UI tail term and the limit-variable tail term are each below $\varepsilon/3$. Choose N so the bounded middle term is below $\varepsilon/3$ for every $n \ge N$. Their sum is below ε. The integrability of X used for its tail follows from the a.s. subsequence and Fatou, as explained above; it must be established before invoking DCT. For that DCT use $|X - \phi_K(X)| \le |X|$ and pointwise convergence to zero.

Original TODO: Prove the Lp version with the correct UI family

For $1 \le p < \infty$, assume $X_n \xrightarrow{\mathbb{P}} X$ and $\{|X_n|^p\}$ is UI. Fatou along an a.s. subsequence gives $\mathbb{E}|X|^p < \infty$. On the event $A_n = \{|X_n - X| > \delta\}$,

$$\mathbb{E}|X_n - X|^p \le \delta^p + 2^{p-1}\left(\mathbb{E}[|X_n|^p \mathbf{1}_{A_n}] + \mathbb{E}[|X|^p \mathbf{1}_{A_n}]\right). \tag{27}$$

Because $\mathbb{P}(A_n) \to 0$, uniform absolute continuity of UI integrals (proved next) makes the first expectation vanish; integrability of $|X|^p$ makes the second vanish. Let $\delta \downarrow 0$. UI of X_n alone does not imply L^p convergence for $p > 1$. For example $X_n = n^{1/p}\mathbf{1}_{(0,1/n)}$ on $(0,1)$ converges to zero in probability and in L^1, is UI, but $\mathbb{E}|X_n|^p = 1$. No finite-moment UI criterion here asserts convergence in L^∞.

Source page(s): 18–21

9 Equivalent characterization of UI

A family $\{X_n\}$ is UI if and only if both conditions hold:

1. $M := \sup_n \mathbb{E}|X_n| < \infty$.
2. For every $\varepsilon > 0$ there exists $\delta > 0$ such that $\mathbb{P}(A) < \delta$ implies $\sup_n \mathbb{E}[|X_n|\mathbf{1}_A] < \varepsilon$ for every measurable event A.

Proof, forward direction. Choose $a > 0$ with $\sup_n \mathbb{E}[|X_n|\mathbf{1}_{\{|X_n|>a\}}] \le 1$. Then

$$\mathbb{E}|X_n| \le a + 1 \quad \text{for all } n. \tag{28}$$

For any measurable A and any $a > 0$, splitting at a gives

$$\mathbb{E}[|X_n|\mathbf{1}_A] \le a\mathbb{P}(A) + \mathbb{E}[|X_n|\mathbf{1}_{\{|X_n|>a\}}]. \tag{29}$$

Choose a so the second term is uniformly below $\varepsilon/2$ and then $\delta = \varepsilon/(2a)$ to bound the first. **Reverse direction.** Markov gives $\sup_n \mathbb{P}(|X_n| > a) \le M/a$. Choose a large enough that this is less than the δ from condition 2. Apply condition 2 to $A = \{|X_n| > a\}$, for each n, to obtain the UI tail bound (reference Proposition 8.3.3).

Original TODO: Review the implication diagram

A uniformly bounded moment of some order greater than one implies UI; UI implies uniformly bounded first moments. Neither reverse implication holds in general. The two-condition criterion above is the necessary-and-sufficient statement. For instance $X_n = n\mathbf{1}_{(0,1/n)}$ has first moments one but is not UI. A single integrable variable repeated as a constant sequence is UI even if it has no finite moment of any order greater than one.

Source page(s): 19

10 Absolute continuity of an integrable function's integral

For integrable X and every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$\mathbb{P}(A) < \delta \quad \Longrightarrow \quad \int_A |X| \, d\mathbb{P} < \varepsilon. \tag{30}$$

By DCT the constant family $X, X, \ldots$ is UI; apply the preceding characterization. Equivalently split the integral at a cutoff and use $\int_A |X| \, d\mathbb{P} \le a\mathbb{P}(A) + \mathbb{E}[|X|\mathbf{1}_{\{|X|>a\}}]$. This is the corollary recorded on page 19 (reference Corollary 8.3.4).

Correction: Small typographical slips in the source and reference

On handwritten page 20 the first equality after splitting the integral must be an inequality when replacing $|X_n|$ by a. The reference's Corollary 8.3.4 ends with $\mathbb{P}(A) < \varepsilon$; its quantified small-probability condition should be $\mathbb{P}(A) < \delta$, as in the statement above. The proof and handwritten version make the intended condition clear.